Indian Statistical Institute

BSDS – Advanced Statistical Methods II

Practice Set 4

Detailed, Self-Explanatory Solutions on Nonlinear Regression

Full problem statements followed by theory, derivations, calculations, and interpretation

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How to use this document. Each problem is organized in three stages: (1) the complete question as stated in the PS4 sheet, (2) the theory and identities needed before beginning the calculation, and (3) a step-by-step derivation that explains why each line follows from the previous one. Numerical answers are rounded only at the final reporting stage where possible.

Source basis. The question statements are transcribed from *PS4 Practice Problems on Non-linear Regression*; explanations follow the notation and methods in the accompanying *NLR short* lecture notes and the PS4 setup sheet. Any genuine modelling caveat is flagged explicitly.

Prepared as a study solution set

Contents

Notation and common tools **2**

1 Problem 1 – Linear or not? **3**

2 Problem 2 – What can the data tell us? **4**

3 Problem 3 – A linear model in disguise **6**

4 Problem 4 – Closed form, but does it exist? **8**

5 Problem 5 – Saturated designs **10**

6 Problem 6 – The three-point method **13**

7 Problem 7 – Profiling out a linear parameter **15**

8 Problem 8 – Logistic regression with a binary covariate **17**

9 Problem 9 – Poisson regression three ways **18**

10 Problem 10 – One step by hand **21**

11 Problem 11 – How fast is Gauss–Newton? **23**

12 Problem 12 – Three tests of one hypothesis **25**

13 Problem 13 – Why profile equals Wald in linear models **27**

14 Problem 14 – Where should we measure? **29**

Consolidated numerical answers **32**

Final checklist for revision **32**

Notation and common tools

The observations are (x_i, y_i), the mean function is $f(x_i, \theta)$, and the residual at parameter value θ is

$$r_i(\theta) = y_i - f(x_i, \theta), \qquad S(\theta) = \sum_{i=1}^{n} r_i(\theta)^2.$$

The nonlinear least-squares estimator (NLSE) is any global minimizer $\widehat{\theta} \in \arg\min_{\theta \in \Theta} S(\theta)$, provided a minimizer is actually attained. The Jacobian is

$$J(\theta) = \left[\frac{\partial f(x_i, \theta)}{\partial \theta_j} \right]_{i,j},$$

and the gradient and Hessian of the sum of squares are

$$\nabla S(\theta) = -2J(\theta)^T r(\theta), \qquad \nabla^2 S(\theta) = 2\left[J^T J - \sum_{i=1}^{n} r_i(\theta) \nabla^2 f(x_i, \theta) \right].$$

For a scalar parameter, write $J_i = f_i'(\theta)$ and $f_i'' = f_i''(\theta)$. Then

$$S'(\theta) = -2 \sum_i J_i r_i, \qquad S''(\theta) = 2\left(\sum_i J_i^2 - \sum_i r_i f_i'' \right).$$

The Gauss–Newton (GN) step uses only the first term in the curvature, $\sum J_i^2$; Newton's step uses the full observed curvature. With independent $N(0, \sigma^2)$ errors, the local large-sample covariance approximation is $\sigma^2 (J^T J)^{-1}$, when the Jacobian has full column rank and the usual regularity conditions hold.

Important distinction used throughout: a graph curved as a function of x may still be linear in its coefficients. A model is intrinsically nonlinear when its unknown parameters enter the mean function nonlinearly; its Jacobian then depends on the unknown parameter values.

1 Problem 1 – Linear or not?

Question (transcribed from PS4)

Problem 1. (Linear or not?) *For each mean function decide: (i) is the model linear in the parameters (use the Jacobian criteria: the model is linear if and only if every $\partial f/\partial\theta_j$ is free of θ)? (ii) Can a transformation make it linear, and under which error structure? (iii) Is there any other trouble?*

(a) $f(x,\theta) = \theta_1 + \theta_2 x + \theta_3 x^2$.

(b) $f(x,\theta) = \theta_1 x^{\theta_2}$, $x > 0$.

(c) $f(x,\theta) = \theta_1 + \theta_2 e^{\theta_3 x}$.

(d) $f(x,\theta) = (\theta_1 + \theta_2 x)^2$.

(e) $f(x,\theta) = \dfrac{\theta_1 x}{\theta_2 + x}$.

Theory needed before solving

A mean function is linear in its parameters if it can be written as $f(x,\theta) = \sum_j \theta_j z_j(x)$, where the regressors $z_j(x)$ do not depend on the parameters. Equivalently, each derivative $\partial f/\partial\theta_j$ is free of θ. A transformation only gives an equivalent statistical model if the transformed errors are treated correctly: taking logs or reciprocals of a response with additive noise generally changes both the error distribution and the variance pattern. A convenient algebraic identity alone does not justify fitting ordinary least squares to the transformed response.

Detailed solution

(a) Quadratic polynomial: linear in the parameters.

$$f = \theta_1 + \theta_2 x + \theta_3 x^2, \qquad \frac{\partial f}{\partial \theta_1} = 1, \quad \frac{\partial f}{\partial \theta_2} = x, \quad \frac{\partial f}{\partial \theta_3} = x^2.$$

Every derivative depends only on x, not on θ. Therefore this is an ordinary linear regression with design columns $1, x, x^2$. The fitted curve may be curved in x, but no nonlinear-regression algorithm is needed. No response transformation is needed. The remaining practical issue is ordinary design-matrix rank: for example, the coefficients cannot all be separately estimated if there are too few distinct x values.

(b) Power law: intrinsically nonlinear, but log-linear under multiplicative errors.

$$f = \theta_1 x^{\theta_2}, \qquad \frac{\partial f}{\partial \theta_1} = x^{\theta_2}, \quad \frac{\partial f}{\partial \theta_2} = \theta_1 x^{\theta_2} \log x.$$

Both derivatives depend on unknown parameters, so the original mean is nonlinear in θ. Suppose the stochastic model is multiplicative, $Y = \theta_1 x^{\theta_2}\eta$, with $\theta_1 > 0$ and $\eta > 0$. Taking logarithms gives

$$\log Y = \log\theta_1 + \theta_2 \log x + \log\eta.$$

This is linear regression in the transformed coefficients $\beta_0 = \log\theta_1$ and $\beta_1 = \theta_2$, with regressor $\log x$. It is not an equivalent additive-error linear model when the original model is $Y = \theta_1 x^{\theta_2} + \varepsilon$: then $\log(Y)$ has a parameter-dependent, generally nonconstant error distribution and may not even be defined for nonpositive observations. The transformed error structure must be plausible. Also, θ_2 cannot be learned if all observed x values equal 1, because then $x^{\theta_2} = 1$ for every θ_2.

(c) Offset exponential: intrinsically nonlinear and conditionally linear.

$$f = \theta_1 + \theta_2 e^{\theta_3 x}, \qquad \frac{\partial f}{\partial \theta_1} = 1, \quad \frac{\partial f}{\partial \theta_2} = e^{\theta_3 x}, \quad \frac{\partial f}{\partial \theta_3} = \theta_2 x e^{\theta_3 x}.$$

The derivatives involving θ_2, θ_3 depend on unknown parameters, so the model is intrinsically nonlinear. However, if θ_3 is fixed, it is linear in θ_1 and θ_2: one can regress y on 1 and $e^{\theta_3 x}$. This motivates profiling the nonlinear parameter θ_3 out. A log transformation such as $\log(Y - \theta_1)$ is only useful if the offset is known and a compatible multiplicative error model is assumed; with an unknown offset and additive error, it is not a simple linearization. If $\theta_2 = 0$, the mean is constant and θ_3 is unidentifiable.

(d) Square of a line: nonlinear in the stated parameters.

$$f = (\theta_1 + \theta_2 x)^2, \quad \frac{\partial f}{\partial \theta_1} = 2(\theta_1 + \theta_2 x), \quad \frac{\partial f}{\partial \theta_2} = 2x(\theta_1 + \theta_2 x).$$

The derivatives depend on θ, so this parameterization is nonlinear. Expanding gives

$$f = \theta_1^2 + 2\theta_1\theta_2 x + \theta_2^2 x^2 = \beta_0 + \beta_1 x + \beta_2 x^2,$$

which has the form of a linear quadratic regression in β, but the coefficients must satisfy the nonlinear restriction $\beta_1^2 = 4\beta_0\beta_2$ and $\beta_0, \beta_2 \geq 0$. Unrestricted polynomial OLS therefore solves a different problem unless those constraints happen to hold. Moreover, (θ_1, θ_2) and $(-\theta_1, -\theta_2)$ give the same curve, so the signs are not jointly identifiable. Taking square roots of Y does not preserve additive error: $\sqrt{(\theta_1 + \theta_2 x)^2 + \varepsilon}$ is not generally $\theta_1 + \theta_2 x$ plus a homoscedastic error.

(e) Michaelis–Menten saturation: intrinsically nonlinear.

$$f = \frac{\theta_1 x}{\theta_2 + x}, \qquad \frac{\partial f}{\partial \theta_1} = \frac{x}{\theta_2 + x}, \quad \frac{\partial f}{\partial \theta_2} = -\frac{\theta_1 x}{(\theta_2 + x)^2}.$$

The Jacobian depends on θ, so the model is nonlinear. Algebraically, when $x \neq 0$ and $f \neq 0$,

$$\frac{1}{f} = \frac{1}{\theta_1} + \frac{\theta_2}{\theta_1}\frac{1}{x}.$$

Thus reciprocal coordinates give a straight-line relation. But if the original observation is $Y = f + \varepsilon$, then $1/Y$ has a nonlinear, heteroscedastic error and can be unstable when Y is near zero. Ordinary least squares on reciprocals generally does not minimize the original-scale sum of squares and is not statistically equivalent to nonlinear least squares. It is exact as a fitting identity only in special cases, such as fitting two group means at two distinct positive x values (see Problem 5). Parameters also require $\theta_2 + x \neq 0$ at every design point; in the usual scientific model $\theta_1, \theta_2 > 0$.

2 Problem 2 – What can the data tell us?

Question (transcribed from PS4)

Problem 2. (What can the data tell us?) *For each model, which parameters (or functions of them) can be estimated? Suggest a fix.*

(a) $f(x, \theta) = \theta_1 e^{\theta_2 + \theta_3 x}$.

(b) $f(x, \theta) = \theta_1(1 - e^{-\theta_2 x})$, *with* $\theta_1, \theta_2 > 0$, *when*

(b1) every observation is taken at the same $x_0 > 0$;

(b2) observations are taken at two distinct values $0 < x_a < x_b$. *Is* θ *now identifiable? Does the NLSE always exist?*

(c) $f(x, \theta) = \theta_1 e^{-\theta_2 x} + \theta_3 e^{-\theta_4 x}$.

(d) $f(x, \theta) = \beta \cos(\theta x)$, *observed at* $x = 1, 2, \ldots, n$.

Theory needed before solving

A parameter vector is structurally identifiable if distinct parameter values cannot generate exactly the same mean function over the design/domain. If two parameter vectors generate the same $f(x, \theta)$ at every possible x, no amount of additional data from that same mean model can distinguish them. This is different from practical identifiability: a parameter may be theoretically identifiable but poorly determined by the available design. Common examples are redundant parameter combinations, label switching in mixtures/sums, symmetries, and nearly flat directions of the objective. A fix is to reparameterize in terms of identifiable combinations, impose scientifically justified constraints, or improve the design.

Detailed solution

(a) Factor the model:

$$f(x, \theta) = \theta_1 e^{\theta_2} e^{\theta_3 x}.$$

Only the combined amplitude $A = \theta_1 e^{\theta_2}$ and the slope/rate θ_3 appear in the mean. For any constant c, replacing θ_1 by $\theta_1 e^{-c}$ and θ_2 by $\theta_2 + c$ leaves A unchanged. Thus θ_1 and θ_2 are not separately identifiable, although their combination $\theta_1 e^{\theta_2}$ can be. Reparameterize as $Ae^{\theta_3 x}$, or fix one of θ_1, θ_2 using external information.

(b) (b1) All observations at a single positive x_0. Every expected response has the same mean

$$m_0 = \theta_1(1 - e^{-\theta_2 x_0}).$$

The data can estimate this one combination, but there is one equation for two unknowns. For any admissible $\theta_2 > 0$, choosing $\theta_1 = m_0/(1 - e^{-\theta_2 x_0})$ yields the same mean (when $m_0 > 0$). The model is not identifiable from this design. Add at least one distinct positive design point and preferably points spanning the approach to the asymptote; alternatively fix one parameter externally.

(b2) Two distinct positive values. Write the two mean values as

$$m_a = \theta_1(1 - e^{-\theta_2 x_a}), \qquad m_b = \theta_1(1 - e^{-\theta_2 x_b}).$$

Their ratio removes the amplitude:

$$\frac{m_a}{m_b} = R(\theta_2) = \frac{1 - e^{-\theta_2 x_a}}{1 - e^{-\theta_2 x_b}}.$$

For $0 < x_a < x_b$ and $\theta_2 > 0$, this ratio varies strictly from x_a/x_b as $\theta_2 \downarrow 0$ to 1 as $\theta_2 \to \infty$. Hence, whenever the observed means are compatible with a finite positive parameter pair, the ratio identifies θ_2 uniquely, and then $\theta_1 = m_a/(1 - e^{-\theta_2 x_a})$. Thus the model is structurally identifiable on this design under the positive-parameter restriction.

However, the NLSE does *not* always exist. The parameter domain $(0, \infty)^2$ is open and noncompact. A continuous objective on such a set need not attain its infimum: minimizing sequences may approach a boundary (for example, $\theta_2 \downarrow 0$ with $\theta_1\theta_2$ approaching a finite limit, producing the limiting line $f(x) \approx (\theta_1\theta_2)x$), or may escape to infinity. Compatibility of exact interpolation and existence of a global least-squares minimizer are separate questions. In practice, inspect parameter bounds and boundary limits, and use scientifically defensible compact constraints or a parameterization that includes the relevant limiting model if appropriate.

(c) The two-exponential model has at least a label-switching symmetry:

$$\theta_1 e^{-\theta_2 x} + \theta_3 e^{-\theta_4 x} = \theta_3 e^{-\theta_4 x} + \theta_1 e^{-\theta_2 x}.$$

Swapping (θ_1, θ_2) with (θ_3, θ_4) leaves the curve unchanged. Consequently, the full labelled vector is not identifiable without an ordering convention. Impose, for example, $\theta_2 < \theta_4$ when scientifically valid. If $\theta_2 = \theta_4$, then the mean reduces to $(\theta_1 + \theta_3)e^{-\theta_2 x}$ and only the amplitude sum is identifiable. If one amplitude is zero, its associated rate is unidentifiable. Even after removing exact symmetries, rates that are close to each other can be practically difficult to distinguish.

(d) Since $\cos(-\theta x) = \cos(\theta x)$, the sign of θ is not identifiable. Furthermore, at integer x,

$$\cos\{(\theta + 2\pi m)x\} = \cos(\theta x + 2\pi m x) = \cos(\theta x)$$

for every integer m. Hence the frequency is aliased modulo 2π, with the additional evenness symmetry. If $\beta = 0$, the mean is zero at every design point and θ has no effect at all. A practical fix is to constrain the frequency to a scientifically justified fundamental interval (typically a non-aliased interval such as $[0, \pi]$ for integer-spaced observations), rule out $\beta = 0$ when appropriate, and use enough distinct sampling points. Frequency identifiability is a design issue, not merely an optimizer issue.

3 Problem 3 – A linear model in disguise

Question (transcribed from PS4)

Problem 3. (A linear model in disguise) *Data:* $x = (1, 2, 3)$, $y = (2.1, 3.9, 6.2)$. *Model:* $Y_i = \theta^3 x_i + \varepsilon_i$, $\theta \in \mathbb{R}$.

(a) Find $\widehat{\theta}$ *in closed form. Does it always exist? Is it unique?*

(b) Show that the Gauss–Newton update is

$$\theta^{(k+1)} = \theta^{(k)} + \frac{\widehat{\beta} - \{\theta^{(k)}\}^3}{3\{\theta^{(k)}\}^2},$$

where $\widehat{\beta}$ *is the least-squares slope of a line through the origin. It is Newton's method for solving which equation?*

(c) Starting from $\theta^{(0)} = 1$*, compute two iterations. What happens from* $\theta^{(0)} = 0$*? From* $\theta^{(0)} = -1$*?*

(d) Show that $\sum_i f_i''(\widehat{\theta}) r_i(\widehat{\theta}) = 0$ *here. What does this imply for the speed of Gauss–Newton (see Problem 11)?*

Theory needed before solving

If a model has the form $f(x, \theta) = g(\theta)x$, and g is one-to-one, first estimate the linear coefficient $\beta = g(\theta)$ by ordinary least squares, then invert g to get θ. For a regression through the origin, $\widehat{\beta} = \sum_i x_i y_i / \sum_i x_i^2$. For a scalar nonlinear least-squares problem, the Gauss–Newton step is

$$\theta_{\text{new}} = \theta + \frac{\sum_i f_i'(\theta) r_i(\theta)}{\sum_i \{f_i'(\theta)\}^2}.$$

Newton's method for solving an equation $h(\theta) = 0$ is $\theta_{\text{new}} = \theta - h(\theta)/h'(\theta)$. Its convergence can fail if $h'(\theta)$ is zero or very small.

Detailed solution

(a) Here $f(x_i, \theta) = \theta^3 x_i$. Set $\beta = \theta^3$; then the model is $Y_i = \beta x_i + \varepsilon_i$, a regression through the origin. Therefore

$$\widehat{\beta} = \frac{\sum_i x_i y_i}{\sum_i x_i^2} = \frac{1(2.1) + 2(3.9) + 3(6.2)}{1^2 + 2^2 + 3^2} = \frac{2.1 + 7.8 + 18.6}{14} = \frac{28.5}{14} = \frac{57}{28}.$$

Since $g(\theta) = \theta^3$ is a one-to-one map from $\mathbb{R}$ onto $\mathbb{R}$, the NLSE is

$$\boxed{\widehat{\theta} = \widehat{\beta}^{1/3} = \left(\frac{57}{28}\right)^{1/3} \approx 1.267376.}$$

It exists and is unique for every real value of $\widehat{\beta}$, because every real number has exactly one real cube root. This closed-form existence statement concerns the global minimizer, not the success of a particular iterative algorithm.

(b) Since $f_i(\theta) = \theta^3 x_i$, we have $J_i = f_i'(\theta) = 3\theta^2 x_i$ and $r_i = y_i - \theta^3 x_i$. Substituting into the scalar GN formula gives

$$\begin{aligned}
\theta_{\text{new}} &= \theta + \frac{\sum_i (3\theta^2 x_i)(y_i - \theta^3 x_i)}{\sum_i (3\theta^2 x_i)^2} \\
&= \theta + \frac{3\theta^2 \{\sum_i x_i y_i - \theta^3 \sum_i x_i^2\}}{9\theta^4 \sum_i x_i^2} \\
&= \theta + \frac{\widehat{\beta} - \theta^3}{3\theta^2}, \qquad \theta \neq 0.
\end{aligned}$$

The final equality uses $\widehat{\beta} = \sum x_i y_i / \sum x_i^2$. This is exactly Newton's method for solving

$$h(\theta) = \theta^3 - \widehat{\beta} = 0,$$

because $h'(\theta) = 3\theta^2$ and $\theta - h/h' = \theta - (\theta^3 - \widehat{\beta})/(3\theta^2)$. It is undefined at $\theta = 0$.

(c) Put $\widehat{\beta} = 57/28 \approx 2.0357143$. Starting at $\theta^{(0)} = 1$:

$$\theta^{(1)} = 1 + \frac{2.0357143 - 1}{3} = 1.3452381.$$

The next iterate is

$$\theta^{(2)} = 1.3452381 + \frac{2.0357143 - (1.3452381)^3}{3(1.3452381)^2} \approx 1.2717960.$$

This moves close to the root 1.2673764 and, under continued iteration from this starting point, converges rapidly.

From $\theta^{(0)} = 0$, the displayed update divides by zero. The model's derivative with respect to θ is also zero at 0, so the local linearization gives no direction. The correct estimator still exists; this particular algorithm is simply unusable at that start.

From $\theta^{(0)} = -1$,

$$\theta^{(1)} = -1 + \frac{2.0357143 - (-1)^3}{3(-1)^2} \approx 0.0119048.$$

The next denominator is proportional to $(0.0119)^2$, so the next step is extremely large and unstable. This illustrates why an exact solution can coexist with poor iterative behavior: Newton/GN updates depend strongly on starting values and can approach a point where their denominator is nearly zero.

(d) For $f_i(\theta) = \theta^3 x_i$, the second derivative is $f_i''(\theta) = 6\theta x_i$. At the estimate,

$$\begin{aligned}\sum_i f_i''(\widehat{\theta}) r_i(\widehat{\theta}) &= 6\widehat{\theta} \sum_i x_i (y_i - \widehat{\theta}^3 x_i) \\ &= 6\widehat{\theta} \left(\sum_i x_i y_i - \widehat{\beta} \sum_i x_i^2 \right) = 0,\end{aligned}$$

by the definition of $\widehat{\beta}$. Problem 11 shows that for a one-parameter model

$$G'(\widehat{\theta}) = \frac{\sum_i f_i''(\widehat{\theta}) r_i(\widehat{\theta})}{\sum_i \{f_i'(\widehat{\theta})\}^2}.$$

Thus $G'(\widehat{\theta}) = 0$. Under the usual smoothness and non-singularity conditions, the leading linear error term vanishes, so GN is locally faster than linear (typically quadratic). This is a local statement and does not fix the bad starts discussed in (c).

4 Problem 4 – Closed form, but does it exist?

Question (transcribed from PS4)

Problem 4. (Closed form, but does it exist?) *Let $Y_i = e^\theta + \varepsilon_i$, $i = 1, \ldots, n$, with ε_i i.i.d. $N(0, \sigma^2)$.*

(a) Find the NLSE $\widehat{\theta}$. When does it exist? Compute the probability that it does not exist when $\theta = 0$, $\sigma = 2$, $n = 4$. (You may use $\Phi(-1) = 0.1587$.)

(b) Is $\widehat{\theta}$ unbiased? Use a second-order expansion to approximate its bias and variance.

(c) Evaluate the approximations for $e^\theta = 5$, $\sigma = 2$, $n = 4$. A simulation with 4×10^6 samples gives bias -0.0212 and standard deviation 0.212. Comment.

(d) Compare your variance with the general formula $\sigma^2 (J^T J)^{-1}$.

(e) Is $e^{\widehat{\theta}}$ an unbiased estimator of e^θ? What does this say about bias and parametrization?

Theory needed before solving

The model's mean is $\mu = e^\theta$, so the parameter space for μ is $(0, \infty)$ even though $\theta \in \mathbb{R}$. For fixed μ, least squares minimizes $\sum_i (y_i - \mu)^2$, whose unconstrained minimizer is $\bar{Y}$. If $\bar{Y} > 0$, it corresponds to a finite $\theta = \log \bar{Y}$. If $\bar{Y} \leq 0$, the objective over $\mu > 0$ approaches its infimum as $\mu \downarrow 0$ but does not attain it; equivalently, $\theta \to -\infty$. This is a concrete example where a closed-form expression is conditional on the minimizer existing.

For a smooth function g, the second-order Taylor expansion at μ is

$$g(\bar{Y}) \approx g(\mu) + g'(\mu)(\bar{Y} - \mu) + \frac{1}{2} g''(\mu)(\bar{Y} - \mu)^2.$$

Writing $v = \mathrm{Var}(\bar{Y}) = \sigma^2/n$, its leading bias approximation is $\frac{1}{2} g''(\mu) v$; the first-order variance approximation is $\{g'(\mu)\}^2 v$. Retaining the quadratic Taylor term also contributes $\frac{1}{2}\{g''(\mu)\}^2 v^2$ to the variance when $\bar{Y}$ is normal, because $\mathrm{Var}\{(\bar{Y} - \mu)^2\} = 2v^2$ and the linear and quadratic terms have zero covariance.

Detailed solution

(a) Let $\mu = e^\theta > 0$. The decomposition

$$\sum_i (Y_i - \mu)^2 = \sum_i (Y_i - \bar{Y})^2 + n(\bar{Y} - \mu)^2$$

shows that the unconstrained minimizer in μ is $\bar{Y}$. Therefore, if $\bar{Y} > 0$,

$$\boxed{\widehat{\theta} = \log \bar{Y}.}$$

If $\bar{Y} \le 0$, no positive μ can equal $\bar{Y}$. The squared-error criterion decreases toward its infimum as $\mu \downarrow 0$, but 0 is not in the parameter space $\mu = e^\theta > 0$. Hence no finite NLSE exists on that event.

Since $\bar{Y} \sim N(e^\theta, \sigma^2/n)$,

$$\Pr(\text{no finite NLSE}) = \Pr(\bar{Y} \le 0) = \Phi\left(-\frac{e^\theta \sqrt{n}}{\sigma}\right).$$

For $\theta = 0$, $e^\theta = 1$, $\sigma = 2$, and $n = 4$, the standard deviation of $\bar{Y}$ is $2/\sqrt{4} = 1$. Thus

$$\Pr(\bar{Y} \le 0) = \Phi\left(\frac{0-1}{1}\right) = \Phi(-1) = \boxed{0.1587}.$$

This probability is substantial: in the specified setting, roughly 16% of samples have no finite least-squares estimate for θ.

(b) Write $\mu = e^\theta$ and $\delta = \bar{Y} - \mu$, so $\mathbb{E}\delta = 0$ and $\mathbb{E}\delta^2 = v = \sigma^2/n$. Use $g(z) = \log z$. Its derivatives are $g'(\mu) = 1/\mu$ and $g''(\mu) = -1/\mu^2$. The second-order expansion is

$$\widehat{\theta} = \log(\mu + \delta) \approx \log \mu + \frac{\delta}{\mu} - \frac{\delta^2}{2\mu^2} = \theta + \frac{\delta}{\mu} - \frac{\delta^2}{2\mu^2}.$$

Taking expectations gives the second-order bias approximation

$$\boxed{\operatorname{Bias}(\widehat{\theta}) \approx -\frac{v}{2\mu^2} = -\frac{\sigma^2}{2ne^{2\theta}}.}$$

It is negative: the concavity of the log pulls the transformed average downward. For the variance, the linear term has variance v/μ^2. The quadratic term has variance

$$\operatorname{Var}\left(\frac{\delta^2}{2\mu^2}\right) = \frac{1}{4\mu^4}\operatorname{Var}(\delta^2) = \frac{2v^2}{4\mu^4} = \frac{v^2}{2\mu^4},$$

where normality gives $\operatorname{Var}(\delta^2) = 2v^2$. Also $\operatorname{Cov}(\delta, \delta^2) = \mathbb{E}(\delta^3) = 0$. Thus the variance of the quadratic approximation is

$$\boxed{\operatorname{Var}(\widehat{\theta}) \approx \frac{v}{\mu^2} + \frac{v^2}{2\mu^4} = \frac{\sigma^2}{ne^{2\theta}} + \frac{\sigma^4}{2n^2 e^{4\theta}}.}$$

The standard first-order delta-method result is the first term alone. These approximations apply where $\bar{Y}$ is positive and near μ; because the log is undefined for $\bar{Y} \le 0$, they are not a substitute for the existence caveat in (a).

(c) Here $\mu = e^\theta = 5$ and $v = \sigma^2/n = 4/4 = 1$. Therefore

$$\text{Bias}(\widehat{\theta}) \approx -\frac{1}{2(5)^2} = -0.0200,$$

and

$$\text{Var}(\widehat{\theta}) \approx \frac{1}{25} + \frac{1}{2(25)^2} = 0.0400 + 0.0008 = 0.0408, \qquad \text{SD}(\widehat{\theta}) \approx \sqrt{0.0408} = 0.2020.$$

The reported simulation bias -0.0212 is very close to -0.0200. Its standard deviation 0.212 is somewhat above the second-order approximation 0.202 (and the first-order approximation 0.200), which is plausible because the Taylor approximation is not exact and the log transformation is asymmetric. At $\mu = 5$, the nonexistence probability is $\Phi(-5)$, about 2.9×10^{-7}, so it is tiny in the simulation; the main difference is finite-sample approximation error, not frequent nonexistence.

(d) Since $f_i(\theta) = e^\theta$, every Jacobian entry is $J_i = e^\theta = \mu$. Consequently

$$J^T J = \sum_{i=1}^{n} J_i^2 = ne^{2\theta} = n\mu^2,$$

and the usual local covariance formula gives

$$\boxed{\sigma^2 (J^T J)^{-1} = \frac{\sigma^2}{ne^{2\theta}} = \frac{v}{\mu^2}.}$$

This agrees exactly with the first-order delta-method variance in (b). The extra $v^2/(2\mu^4)$ is a higher-order term from retaining the quadratic Taylor term, and so does not appear in the standard first-order asymptotic formula.

(e) On the event $\bar{Y} > 0$ where the finite estimate exists,

$$e^{\widehat{\theta}} = e^{\log \bar{Y}} = \bar{Y}.$$

The sample mean is unbiased for $e^\theta = \mu$: $\mathbb{E}\bar{Y} = \mu$. Thus the back-transformed estimate is algebraically the unbiased sample mean whenever the log-parameter estimate exists. There is a formal qualification: under the stated parameter space $\mu > 0$, $\widehat{\theta}$ and hence $e^{\widehat{\theta}}$ are undefined on $\bar{Y} \le 0$, so it is not a globally defined estimator without an extra convention. If one instead closes the mean parameter space to $\mu \ge 0$, the constrained estimator is $\widehat{\mu} = \max(0, \bar{Y})$, which is not exactly unbiased. The lesson is that bias is parameterization-dependent: a biased estimate of $\theta = \log \mu$ can transform back to an unbiased estimate of μ, but existence and the estimator's definition must be stated carefully.

5 Problem 5 – Saturated designs

Question (transcribed from PS4)

Problem 5. (Saturated designs)

(a) Suppose the design has exactly p distinct values $x_1, \ldots, x_p$, with n_k observations and mean $\bar{y}_k$ at x_k. Show that

$$S(\theta) = \sum_{k=1}^{p} \sum_{i=1}^{n_k} (y_{ki} - \bar{y}_k)^2 + \sum_{k=1}^{p} n_k \{\bar{y}_k - f(x_k, \theta)\}^2,$$

and deduce: if $f(x_k, \theta^*) = \bar{y}_k$ for all k, then θ^* is an NLSE.

(b) $f(x, \theta) = \theta_1 e^{-\theta_2 x}$, with observations at $x = 0$ and $x = c$ only. Find the NLSE in closed form. When does it exist? Evaluate it for $\bar{y}_0 = 100$, $\bar{y}_c = 40$, $c = 2$ hours.

(c) Michaelis–Menten model $f(x; V_m, K) = V_m x/(K + x)$, using only the Puromycin (treated) runs at $x = 0.02$ (rates $76, 47$) and $x = 1.10$ (rates $207, 200$). Find $\widehat{V}_m$ and $\widehat{K}$ in closed form.

(d) The Lineweaver–Burk method (OLS of $1/y$ on $1/x$) generally gives biased estimates of V_m and K. Why is it exact here?

Theory needed before solving

For each design point x_k, write the residual as $y_{ki} - f_k = (y_{ki} - \bar{y}_k) + (\bar{y}_k - f_k)$. Squaring and summing kills the cross term because $\sum_i (y_{ki} - \bar{y}_k) = 0$. This splits the residual sum of squares into (i) within-group variation, which the fitted mean cannot remove, and (ii) a nonnegative penalty for missing each group mean. If a model with p parameters can interpolate the means at exactly p distinct design points, it reaches the smallest possible value of S. Such a design is called saturated for those group means.

Detailed solution

(a) At design point x_k, let $f_k = f(x_k, \theta)$. Then

$$y_{ki} - f_k = (y_{ki} - \bar{y}_k) + (\bar{y}_k - f_k).$$

Squaring and summing over i,

$$\sum_{i=1}^{n_k} (y_{ki} - f_k)^2 = \sum_i (y_{ki} - \bar{y}_k)^2 + 2(\bar{y}_k - f_k) \sum_i (y_{ki} - \bar{y}_k) + n_k (\bar{y}_k - f_k)^2.$$

But $\sum_i (y_{ki} - \bar{y}_k) = 0$ by the definition of the sample mean. Therefore

$$\sum_i (y_{ki} - f_k)^2 = \sum_i (y_{ki} - \bar{y}_k)^2 + n_k (\bar{y}_k - f_k)^2.$$

Summing this equality over $k = 1, \ldots, p$ yields the required decomposition. The first sum does not depend on θ; the second is nonnegative. If $f(x_k, \theta^*) = \bar{y}_k$ for every k, every term in the second sum is zero, so no other parameter can produce a smaller objective. Hence θ^* is a global NLSE.

(b) At $x = 0$, $f(0) = \theta_1$, so exact interpolation of the group mean implies

$$\widehat{\theta}_1 = \bar{y}_0.$$

At $x = c$, interpolation requires $\bar{y}_c = \bar{y}_0 e^{-\theta_2 c}$. If the ratio is positive and $\bar{y}_0 \neq 0$, solve for θ_2:

$$e^{-\widehat{\theta}_2 c} = \frac{\bar{y}_c}{\bar{y}_0}, \qquad \boxed{\widehat{\theta}_2 = -\frac{1}{c} \log\left(\frac{\bar{y}_c}{\bar{y}_0}\right) = \frac{1}{c} \log\left(\frac{\bar{y}_0}{\bar{y}_c}\right).}$$

Under the unconstrained real parameterization written in the question, this exact interpolator exists when $\bar{y}_0 \neq 0$ and $\bar{y}_c / \bar{y}_0 > 0$. If both means are zero, $\theta_1 = 0$ matches both and θ_2 is arbitrary. If $\bar{y}_0 \neq 0$ but $\bar{y}_c = 0$, no finite θ_2 interpolates the second mean, because an exponential is never exactly zero; likewise opposite-signed means cannot be interpolated. (The existence of a global minimizer in a non-interpolating edge case is a separate optimization question.)

In the usual physical decay model, one additionally imposes $\theta_1 > 0$ and $\theta_2 > 0$, so the exact interpolating estimate requires $0 < \bar{y}_c < \bar{y}_0$. For the supplied numbers,

$$\widehat{\theta}_1 = 100, \qquad \widehat{\theta}_2 = \frac{1}{2}\log\frac{100}{40} = \frac{1}{2}\log(2.5) \approx 0.458145 \text{ per hour.}$$

The fitted curve is $\widehat{f}(x) = 100e^{-0.458145x}$, and it predicts 40 at $x = 2$ hours, as required.

(c) First compute the two group means:

$$\bar{y}_a = \frac{76+47}{2} = 61.5, \qquad \bar{y}_b = \frac{207+200}{2} = 203.5,$$

where $a = 0.02$ and $b = 1.10$. Saturation says the two-parameter NLS fit interpolates these means. Thus

$$\bar{y}_a = \frac{V_m a}{K+a}, \qquad \bar{y}_b = \frac{V_m b}{K+b}.$$

Rearrange each equation as $V_m = \bar{y}_a(K+a)/a = \bar{y}_b(K+b)/b$. Equating these expressions gives

$$b\bar{y}_a(K+a) = a\bar{y}_b(K+b).$$

Collect the terms involving K:

$$K(b\bar{y}_a - a\bar{y}_b) = ab(\bar{y}_b - \bar{y}_a).$$

Therefore

$$\boxed{\widehat{K} = \frac{ab(\bar{y}_b - \bar{y}_a)}{b\bar{y}_a - a\bar{y}_b}}, \qquad \boxed{\widehat{V}_m = \frac{\bar{y}_a(\widehat{K}+a)}{a}.}$$

Substitution yields

$$\widehat{K} = \frac{(0.02)(1.10)(203.5-61.5)}{(1.10)(61.5)-(0.02)(203.5)} \approx 0.049135,$$

$$\widehat{V}_m = \frac{61.5(0.04913495+0.02)}{0.02} \approx 212.590.$$

Hence, using exactly the two group means specified, $\boxed{\widehat{V}_m \approx 212.59,\ \widehat{K} \approx 0.04913}$. These are the exact two-point interpolating NLS estimates for this reduced design; they need not equal estimates obtained from all Puromycin observations in the full data set.

(d) With two positive group means $(a, \bar{y}_a)$ and $(b, \bar{y}_b)$, the reciprocal identity is

$$\frac{1}{f(x)} = \frac{1}{V_m} + \frac{K}{V_m}\frac{1}{x}.$$

The two transformed mean points determine a unique straight line, provided the resulting parameters are admissible. There are two distinct group means and two model parameters, so the line fits both transformed means exactly. Its intercept and slope then define the same Michaelis–Menten curve that passes through the two original means. Thus the method is algebraically exact for this saturated two-mean design.

Important interpretation. This does not mean the reciprocal estimator is generally statistically unbiased. Nor is it automatically exact if one runs OLS on all four raw reciprocal observations separately: within a repeated-x group, $\overline{(1/Y)}$ is not generally $1/\bar{Y}$. For these four raw rates, ungrouped reciprocal OLS gives approximately $\widehat{V}_m = 213.33$ and $\widehat{K} = 0.05346$, not the original-scale NLS values from (c). Thus the phrase "exact here" is correct only when the analysis is explicitly performed on the two original-scale group means and then transformed; it is not true for every implementation of Lineweaver–Burk on the four individual observations. On the original additive-error scale, reciprocal fitting to all raw values changes the weighting and error structure.

6 Problem 6 – The three-point method

Question (transcribed from PS4)

Problem 6. (The three-point method) *Asymptotic regression:* $y(x) = \alpha + (y_0 - \alpha)e^{-kx}$, $k > 0$. *There are three equally spaced design points* $x, x+h, x+2h$, *with means* $\bar{y}_1, \bar{y}_2, \bar{y}_3$.

(a) Show that $y(x+h) - \alpha = \rho\{y(x) - \alpha\}$ *with* $\rho = e^{-kh}$. *Deduce the closed forms*

$$\widehat{\rho} = \frac{\bar{y}_3 - \bar{y}_2}{\bar{y}_2 - \bar{y}_1}, \quad \widehat{k} = -\frac{\log \widehat{\rho}}{h}, \quad \widehat{\alpha} = \frac{\bar{y}_1\bar{y}_3 - \bar{y}_2^2}{\bar{y}_1 + \bar{y}_3 - 2\bar{y}_2}.$$

Why is this the NLSE?

(b) A cup of coffee reads $90, 70, 56°C$ *at* $t = 0, 5, 10$ *minutes. Estimate the room temperature and the cooling rate, and predict the reading at* $t = 15$.

(c) When does the solution exist? Try the biochemical oxygen demand (BOD) data at days $1, 2, 3$ $(8.3, 10.3, 19.0)$ *and at days* $3, 4, 5$ $(19.0, 16.0, 15.6)$.

(d) Why is the method still useful when there are many data points?

Theory needed before solving

For equally spaced observations from an exponential approach to an asymptote, define $\rho = e^{-kh}$. Since $k > 0$ and $h > 0$, $0 < \rho < 1$. Each step toward the asymptote shrinks by the same factor:

$$y(x+h) - \alpha = \rho\{y(x) - \alpha\}.$$

With three means, there are three unknowns (asymptote, rate, and amplitude/initial level), so if the three-point equations can be solved with admissible parameters, the curve interpolates all three means. By the saturated-design decomposition in Problem 5, that interpolation minimizes the sum of squares for those three design means.

Detailed solution

(a) Starting with the model,

$$y(x) - \alpha = (y_0 - \alpha)e^{-kx}.$$

At the next point,

$$\begin{aligned} y(x+h) - \alpha &= (y_0 - \alpha)e^{-k(x+h)} \\ &= e^{-kh}(y_0 - \alpha)e^{-kx} \\ &= e^{-kh}\{y(x) - \alpha\}. \end{aligned}$$

Therefore $\rho = e^{-kh}$ and $y_{j+1} - \alpha = \rho(y_j - \alpha)$. For the three means,

$$\bar{y}_2 - \alpha = \rho(\bar{y}_1 - \alpha), \qquad \bar{y}_3 - \alpha = \rho(\bar{y}_2 - \alpha).$$

Subtract the first equation from the second relation in difference form:

$$\bar{y}_2 - \bar{y}_1 = (\rho - 1)(\bar{y}_1 - \alpha), \qquad \bar{y}_3 - \bar{y}_2 = (\rho - 1)(\bar{y}_2 - \alpha).$$

Taking the ratio, when the denominator is nonzero,

$$\boxed{\widehat{\rho} = \frac{\bar{y}_3 - \bar{y}_2}{\bar{y}_2 - \bar{y}_1}.}$$

Then $\rho = e^{-kh}$ implies $\log \rho = -kh$, giving $\boxed{\widehat{k} = -\log(\widehat{\rho})/h}$. To obtain α, use $\bar{y}_2 - \alpha = \rho(\bar{y}_1 - \alpha)$ and eliminate ρ using the difference ratio. A direct rearrangement yields

$$\boxed{\widehat{\alpha} = \frac{\bar{y}_1\bar{y}_3 - \bar{y}_2^2}{\bar{y}_1 + \bar{y}_3 - 2\bar{y}_2}.}$$

The solution is admissible for the stated model only when the expressions are defined and $0 < \widehat{\rho} < 1$, which makes $\widehat{k} > 0$. The implied amplitude/initial value is then determined from any of the three interpolation equations.

Why is it the NLSE? The model has three free parameters and there are exactly three distinct design points. When the above solution exists, the model passes through the three group means. By Problem 5(a), this removes the nonnegative between-group term from the residual sum of squares, achieving a global minimum.

(b) The observations are equally spaced by $h = 5$ minutes, with $\bar{y}_1 = 90$, $\bar{y}_2 = 70$, and $\bar{y}_3 = 56$. Hence

$$\widehat{\rho} = \frac{56 - 70}{70 - 90} = \frac{-14}{-20} = 0.7.$$

Since $0 < 0.7 < 1$, the fitted cooling rate is admissible:

$$\widehat{k} = -\frac{\log(0.7)}{5} \approx 0.071334 \text{ min}^{-1}.$$

The room temperature (the asymptote) is

$$\widehat{\alpha} = \frac{90(56) - 70^2}{90 + 56 - 2(70)} = \frac{5040 - 4900}{6} = \frac{70}{3} \approx 23.333°\text{C}.$$

The fitted expression anchored at $t = 0$ is

$$\widehat{y}(t) = \widehat{\alpha} + (90 - \widehat{\alpha})e^{-\widehat{k}t}.$$

At $t = 15$, use the three-step recurrence, $\widehat{y}(15) - \widehat{\alpha} = \widehat{\rho}^3(90 - \widehat{\alpha})$:

$$\widehat{y}(15) = \frac{70}{3} + \left(90 - \frac{70}{3}\right)(0.7)^3 = 23.333 + 66.667(0.343) = \boxed{46.2°\text{C}}.$$

(c) The algebraic formula requires a nonzero first difference $\bar{y}_2 - \bar{y}_1$ and a nonzero denominator $\bar{y}_1 + \bar{y}_3 - 2\bar{y}_2$. Most importantly, a finite rate $k > 0$ requires $0 < \widehat{\rho} < 1$. In words, the two successive increments must have the same sign, and the second increment must be strictly smaller in magnitude than the first. The data must display a monotone movement toward an asymptote with shrinking steps.

For BOD at days $1, 2, 3$: $(8.3, 10.3, 19.0)$,

$$\widehat{\rho} = \frac{19.0 - 10.3}{10.3 - 8.3} = \frac{8.7}{2.0} = 4.35 > 1.$$

Then $\widehat{k} = -\log(4.35) < 0$, contradicting $k > 0$. These three means do not fit the required positive-rate asymptotic model. Also $\widehat{\alpha} \approx 7.703$ from the algebraic expression, but the rate constraint alone disqualifies this solution.

For BOD at days $3, 4, 5$: $(19.0, 16.0, 15.6)$,

$$\widehat{\rho} = \frac{15.6 - 16.0}{16.0 - 19.0} = \frac{-0.4}{-3.0} = 0.133333\ldots,$$

which is in $(0, 1)$. Thus

$$\widehat{k} = -\log(0.133333\ldots) \approx 2.01490 \text{ day}^{-1},$$

$$\widehat{\alpha} = \frac{19.0(15.6) - 16.0^2}{19.0 + 15.6 - 32.0} = \frac{40.4}{2.6} \approx \boxed{15.5385}.$$

This is a valid declining-to-asymptote fit to these three means. It does not say that every BOD sequence must decline; it diagnoses whether this particular three-point window matches the specified asymptotic pattern.

(d) With many data points, the three-point method is useful for constructing starting values for nonlinear least squares. Select three suitably spaced observations or group means that appear to follow the asymptotic regime, compute rough α and k, then fit all observations by NLS. It can also serve as a quick diagnostic: if many triples give incompatible values or violate $0 < \rho < 1$, the asymptotic model may be inappropriate over the chosen range. For noisy data, the three-point estimates themselves may be unstable, so they are generally initialization/diagnostic tools rather than a replacement for fitting all observations.

7 Problem 7 – Profiling out a linear parameter

Question (transcribed from PS4)

Problem 7. (Profiling out a linear parameter) *Consider $Y_i = \theta_1 g_i(\theta_2) + \varepsilon_i$ with $g_i(\theta_2) = 1 - e^{-\theta_2 x_i}$ (the BOD model). Write $g(\theta_2) = (g_1, \ldots, g_n)^T$ and $y = (y_1, \ldots, y_n)^T$.*

(a) For fixed θ_2, find $\widehat{\theta}_1(\theta_2)$ minimizing $S(\theta_1, \theta_2)$.

(b) Show that the profile sum of squares is

$$\widetilde{S}(\theta_2) = S(\widehat{\theta}_1(\theta_2), \theta_2) = \|y\|^2 - \frac{(y^T g)^2}{\|g\|^2} = \|y\|^2 \sin^2 \alpha(\theta_2),$$

where $\alpha(\theta_2)$ is the angle between y and $g(\theta_2)$. Interpret $\widehat{\theta}_2$ geometrically.

(c) Show that $\dfrac{d\widetilde{S}}{d\theta_2} = \left.\dfrac{\partial S}{\partial \theta_2}\right|_{\theta_1 = \widehat{\theta}_1(\theta_2)}$, and conclude that minimizing $\widetilde{S}$ gives the NLSE.

(d) Why is this useful in practice?

Theory needed before solving

Profiling, also called variable projection, minimizes analytically over a parameter that enters linearly while leaving a lower-dimensional numerical problem in the nonlinear parameters. For a fixed vector g, minimizing $\|y - ag\|^2$ over a is an ordinary least-squares projection of y onto the one-dimensional span of g. The residual is orthogonal to g. The envelope theorem says that when a nuisance parameter is already optimized for each value of the remaining parameter, the derivative of the profile objective is its partial derivative with respect to the remaining parameter: the term from the derivative of the optimized nuisance parameter vanishes because its own first-order condition is zero.

Detailed solution

(a) For fixed θ_2, abbreviate $g = g(\theta_2)$. Then

$$S(\theta_1, \theta_2) = \|y - \theta_1 g\|^2 = y^T y - 2\theta_1 y^T g + \theta_1^2 g^T g.$$

Differentiate with respect to θ_1 and set the derivative to zero:

$$\frac{\partial S}{\partial \theta_1} = -2y^T g + 2\theta_1 g^T g = 0.$$

Provided $g^T g > 0$, solve to get

$$\boxed{\widehat{\theta}_1(\theta_2) = \frac{y^T g(\theta_2)}{g(\theta_2)^T g(\theta_2)}.}$$

This is exactly the OLS coefficient for regressing y on the single regressor vector g through the origin.

(b) Substitute the minimizer into the quadratic from (a):

$$\begin{aligned}\widetilde{S}(\theta_2) &= y^T y - 2\frac{y^T g}{g^T g} y^T g + \frac{(y^T g)^2}{(g^T g)^2} g^T g \\ &= \|y\|^2 - \frac{(y^T g)^2}{\|g\|^2}.\end{aligned}$$

By definition of the angle α between two nonzero vectors,

$$\cos\alpha = \frac{y^T g}{\|y\|\|g\|}, \quad \frac{(y^T g)^2}{\|g\|^2} = \|y\|^2 \cos^2\alpha.$$

Using $1 - \cos^2\alpha = \sin^2\alpha$ gives

$$\boxed{\widetilde{S}(\theta_2) = \|y\|^2 \sin^2\alpha(\theta_2).}$$

Geometrically, the best θ_2 is the value for which the direction of $g(\theta_2)$ comes as close as possible to the direction of y (or its opposite if signs are unconstrained). The amplitude θ_1 then scales that direction to the orthogonal projection of y onto the line spanned by g.

(c) By the chain rule,

$$\frac{d\widetilde{S}}{d\theta_2} = \frac{\partial S}{\partial \theta_2} + \frac{\partial S}{\partial \theta_1}\frac{d\widehat{\theta}_1}{d\theta_2},$$

where both partial derivatives are evaluated at $\theta_1 = \widehat{\theta}_1(\theta_2)$. From the normal equation in (a),

$$\frac{\partial S}{\partial \theta_1}\Big|_{\theta_1 = \widehat{\theta}_1(\theta_2)} = 0.$$

Thus the second term vanishes, proving

$$\boxed{\frac{d\widetilde{S}}{d\theta_2} = \frac{\partial S}{\partial \theta_2}\Big|_{\theta_1 = \widehat{\theta}_1(\theta_2)}.}$$

For every fixed θ_2, the chosen $\widehat{\theta}_1(\theta_2)$ already minimizes S over θ_1. Therefore minimizing the profile $\widetilde{S}$ over θ_2 minimizes S jointly over (θ_1, θ_2); at a global minimizer, $\widehat{\theta}_1$ is recovered from the formula in (a).

(d) The original optimization searches over two parameters, but profiling reduces it to a one-dimensional optimization over θ_2. A one-dimensional objective is easier to visualize, grid-search, bracket, and diagnose for multiple minima. It also avoids numerical instability associated with treating an easy linear parameter as if it were fully nonlinear. The same idea extends to many nonlinear models with linear amplitudes, intercepts, or asymptotes; it often produces better starting values and more reliable optimization.

8 Problem 8 – Logistic regression with a binary covariate

Question (transcribed from PS4)

Problem 8. (Logistic regression with a binary covariate) *The Challenger O-ring data, with launches split at 65°F:*

	flights	*O-rings at risk*	*O-rings distressed*
cold ($< 65°$F)	4	24	5
warm ($\geq 65°$F)	19	114	4

Model: $\operatorname{logit}\pi = \beta_0 + \beta_1 \cdot$ *cold, with binomial counts.*

(a) Show that the MLE has a closed form: $\widehat{\pi}_g = k_g/m_g$ *in each group* g*, where* k_g *is the number distressed and* m_g *is the number at risk. Find* $\widehat{\beta}_0$ *and* $\widehat{\beta}_1$*, and interpret* $e^{\widehat{\beta}_1}$*.*

(b) Show that the estimated variance of $\widehat{\beta}_1$ *is* $1/k_c + 1/(m_c - k_c) + 1/k_w + 1/(m_w - k_w)$*. Give a 95% interval for the odds ratio (*$z_{0.975} = 1.96$*).*

(c) In what sense is this a saturated design, as in Problem 5?

(d) With temperature as a continuous covariate, the same model gives $\widehat{\beta}_1 = -0.1156$ *per °F. What is lost by dichotomizing temperature?*

Theory needed before solving

For a binomial group with k successes in m trials and success probability π, the log-likelihood (up to constants) is $\ell(\pi) = k\log\pi + (m-k)\log(1-\pi)$. Differentiating shows the MLE is $\widehat{\pi} = k/m$. Under a two-group logistic model with a binary indicator, β_0 is the log-odds in the reference group and β_1 is the difference in log-odds between the exposed and reference groups. Thus e^{β_1} is the odds ratio. For independent binomial groups, the large-sample estimated variance of a log-odds ratio is the sum of the four reciprocal cell counts in the corresponding 2×2 table.

Detailed solution

(a) Label warm as the reference group (cold = 0) and cold as the indicator group (cold = 1). For a single group,

$$\ell(\pi) = k\log\pi + (m-k)\log(1-\pi) + C,$$

so

$$\ell'(\pi) = \frac{k}{\pi} - \frac{m-k}{1-\pi} = 0 \quad \Longrightarrow \quad k(1-\pi) = (m-k)\pi,$$

which simplifies to $\widehat{\pi} = k/m$. Hence

$$\widehat{\pi}_w = \frac{4}{114} = 0.0350877, \qquad \widehat{\pi}_c = \frac{5}{24} = 0.2083333.$$

The logistic form is $\operatorname{logit}\pi = \beta_0 + \beta_1\,\text{cold}$. In the warm group,

$$\widehat{\beta}_0 = \operatorname{logit}(\widehat{\pi}_w) = \log\left(\frac{4}{114-4}\right) = \log\left(\frac{4}{110}\right) \approx -3.314187.$$

In the cold group,

$$\widehat{\beta}_0 + \widehat{\beta}_1 = \operatorname{logit}(\widehat{\pi}_c) = \log\left(\frac{5}{24-5}\right) = \log\left(\frac{5}{19}\right).$$

Subtracting gives

$$\widehat{\beta}_1 = \log\left(\frac{5}{19}\right) - \log\left(\frac{4}{110}\right) = \log\left(\frac{5/19}{4/110}\right) \approx 1.979185.$$

Therefore

$$\boxed{e^{\widehat{\beta}_1} = \frac{5/19}{4/110} \approx 7.23684.}$$

The estimated odds of O-ring distress in the cold group are about 7.24 times the estimated odds in the warm group. This is an odds ratio, not a risk ratio or a statement that the probability itself is multiplied by 7.24.

(b) The estimated variance is

$$\begin{aligned}\widehat{\mathrm{Var}}(\widehat{\beta}_1) &= \frac{1}{5} + \frac{1}{24-5} + \frac{1}{4} + \frac{1}{114-4} \\ &= 0.2 + 0.0526316 + 0.25 + 0.0090909 \\ &= \boxed{0.5117225}.\end{aligned}$$

Thus $\mathrm{se}(\widehat{\beta}_1) = \sqrt{0.5117225} \approx 0.715348$. Form the Wald interval on the log-odds-ratio scale:

$$\widehat{\beta}_1 \pm 1.96\ \mathrm{se}(\widehat{\beta}_1) = 1.979185 \pm 1.96(0.715348) \approx (0.5769, 3.3815).$$

Exponentiating both endpoints gives a 95% confidence interval for the odds ratio:

$$\boxed{\exp(0.5769, 3.3815) \approx (1.78, 29.41).}$$

The interval is wide, reflecting the small number of distress events (five cold, four warm). It lies above one under this Wald approximation, but its width signals substantial uncertainty about the magnitude.

(c) The binary covariate has exactly two distinct design values, cold = 0 and cold = 1, and the model has two parameters, β_0 and β_1. It can fit the two observed group proportions exactly: $\widehat{\pi}_w = 4/114$ and $\widehat{\pi}_c = 5/24$. In that sense it is saturated for the two group probabilities, analogous to a two-point saturated mean design in Problem 5. The likelihood still has binomial sampling variation within each group; saturation means the model does not impose an additional relationship between the two group probabilities beyond their separate estimates.

(d) Dichotomizing temperature discards information about actual temperatures within each category. Launches just below and far below 65°F are treated as identical; likewise for warm launches. This generally loses statistical power/precision, hides within-group trends, makes results depend on the selected cut-off, and prevents estimating a per-degree change or detecting nonlinear temperature patterns. The continuous estimate -0.1156 per °F describes change in log-odds for each one-degree increase (under the continuous linear-logit model); it is not directly comparable numerically with the binary coefficient, which compares two groups.

9 Problem 9 – Poisson regression three ways

Question (transcribed from PS4)

Problem 9. (Poisson regression three ways) $Y_i \sim \mathrm{Poisson}(\mu_i)$ *independently, with* $\log \mu_i = \eta_i = x_i^T \beta$.

(a) Write down the log-likelihood, the score, the observed information $-\nabla^2\ell$, and the expected information. Show that the last two coincide.

(b) Show that the Newton (equivalently, Fisher scoring) update is a weighted least-squares fit of a working response z_i on x_i, with weights w_i. Find z_i and w_i.

(c) Data: $x = (0, 1, 2)$, $y = (2, 3, 6)$, model $\log\mu = \beta_0 + \beta_1 x$. Starting from $\beta^{(0)} = (\log\bar{y}, 0)$, compute $\beta^{(1)}$ by hand.

(d) Why can there be no spurious local maxima here, unlike in nonlinear least squares?

(e) Show that at the MLE, $\sum_i y_i = \sum_i \widehat{\mu}_i$.

Theory needed before solving

For a Poisson response, $\Pr(Y_i = y_i) = e^{-\mu_i}\mu_i^{y_i}/y_i!$. The canonical log link is $\eta_i = \log\mu_i = x_i^T\beta$, so $\mu_i = e^{x_i^T\beta}$. The score is $X^T(y-\mu)$ and the Hessian is $-X^T\,\mathrm{diag}(\mu)X$. Because μ depends only on the current parameters and the observed Hessian contains no random Y_i term, the observed and expected information coincide. The negative log-likelihood is convex (equivalently the log-likelihood is concave), preventing spurious local maxima; strict concavity requires a full-rank design and positive means.

Detailed solution

(a) Independence gives the likelihood

$$L(\beta) = \prod_i \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}.$$

Taking logarithms and using $\log\mu_i = x_i^T\beta$,

$$\ell(\beta) = \sum_i \{y_i x_i^T\beta - e^{x_i^T\beta} - \log(y_i!)\}.$$

Differentiate with respect to β:

$$\boxed{\nabla\ell(\beta) = \sum_i x_i(y_i - \mu_i) = X^T(y-\mu).}$$

Since $\partial\mu_i/\partial\beta = \mu_i x_i$, the Hessian is

$$\nabla^2\ell(\beta) = -\sum_i \mu_i x_i x_i^T = -X^T\,\mathrm{diag}(\mu_i)X.$$

Therefore observed information is

$$\boxed{\mathcal{I}_{\mathrm{obs}}(\beta) = -\nabla^2\ell = X^T\,\mathrm{diag}(\mu_i)X.}$$

It is a deterministic function of β and the fixed design, so taking expectation does not change it:

$$\boxed{\mathcal{I}_{\mathrm{exp}}(\beta) = \mathbb{E}\{\mathcal{I}_{\mathrm{obs}}(\beta)\} = X^T\,\mathrm{diag}(\mu_i)X.}$$

Thus Newton and Fisher scoring have the same update in this model.

(b) Let $W = \mathrm{diag}(\mu_i)$. The Newton/Fisher-scoring step is

$$\beta^{\mathrm{new}} = \beta + (X^TWX)^{-1}X^T(y-\mu).$$

Define the working response

$$z_i = \eta_i + \frac{y_i - \mu_i}{\mu_i}, \qquad w_i = \mu_i.$$

In vector form, $z = X\beta + W^{-1}(y - \mu)$. Then

$$X^T W z = X^T W X \beta + X^T (y - \mu),$$

so the weighted least-squares solution is

$$(X^T W X)^{-1} X^T W z = \beta + (X^T W X)^{-1} X^T (y - \mu),$$

which is precisely the Newton/Fisher-scoring update. Therefore the iteratively reweighted least-squares (IRLS) weights and responses are

$$\boxed{w_i = \mu_i, \qquad z_i = \eta_i + (y_i - \mu_i)/\mu_i.}$$

All quantities on the right are evaluated at the current iterate.

(c) First compute $\bar{y} = (2 + 3 + 6)/3 = 11/3$, so

$$\beta^{(0)} = (\log(11/3), 0), \quad \eta_i^{(0)} = \log(11/3), \quad \mu_i^{(0)} = 11/3.$$

Write $\eta_0 = \log(11/3)$ and $\mu_0 = 11/3$. The working responses are

$$\begin{aligned} z_0 &= \eta_0 + \frac{2 - 11/3}{11/3} = \eta_0 - \frac{5}{11}, \\ z_1 &= \eta_0 + \frac{3 - 11/3}{11/3} = \eta_0 - \frac{2}{11}, \\ z_2 &= \eta_0 + \frac{6 - 11/3}{11/3} = \eta_0 + \frac{7}{11}. \end{aligned}$$

All three weights equal $11/3$, so weighted least squares is just ordinary least squares of z on an intercept and $x = (0, 1, 2)$. The slope is

$$\widehat{\beta}_1^{(1)} = \frac{z_2 - z_0}{2} = \frac{(7/11) - (-5/11)}{2} = \frac{6}{11} \approx 0.545455.$$

Because $\bar{x} = 1$ and the mean of the three z_i is η_0, the intercept is $\widehat{\beta}_0^{(1)} = \bar{z} - \widehat{\beta}_1^{(1)} \bar{x} = \eta_0 - 6/11$. Hence

$$\boxed{\beta^{(1)} = \left(\log \frac{11}{3} - \frac{6}{11}, \frac{6}{11}\right) \approx (0.75383, 0.54545).}$$

(d) The log-likelihood has Hessian $-X^T W X$, which is negative semidefinite at every β. Thus ℓ is concave, so any finite stationary point is a global maximum; if X has full column rank, the Hessian is negative definite and the log-likelihood is strictly concave, so the finite MLE is unique. Nonlinear least-squares objectives can be nonconvex because their Hessian has an additional residual-curvature term; that term can create multiple local minima. A caveat: concavity does not itself guarantee that a finite MLE exists for every data/design configuration, but it rules out spurious nonglobal local maxima when a finite maximum exists.

(e) The score component for the intercept (whose design value is 1 for every observation) is

$$\frac{\partial \ell}{\partial \beta_0} = \sum_i (y_i - \mu_i).$$

At an interior finite MLE, the score is zero, so

$$\sum_i (y_i - \widehat{\mu}_i) = 0 \quad \Longrightarrow \quad \boxed{\sum_i y_i = \sum_i \widehat{\mu}_i.}$$

This is a consequence of including an intercept. It says the fitted total count matches the observed total count at the MLE.

10 Problem 10 – One step by hand

Question (transcribed from PS4)

Problem 10. (One step by hand) *Data:* $x = (1, 2)$, $y = (0.62, 0.18)$. *Model:* $f(x, \theta) = e^{-\theta x}$. *Start:* $\theta^{(0)} = 1$.

(a) Compute f_i, r_i, J_i *and* f_i'' *at* $\theta^{(0)}$.

(b) Compute one Gauss–Newton step $\theta^{(1)}_{\mathrm{GN}}$.

(c) Compute one Newton step $\theta^{(1)}_{N}$.

(d) The minimizer is $\widehat{\theta} = 0.6489$. *Evaluate* S *at* $\theta^{(0)}$, $\theta^{(1)}_{\mathrm{GN}}$, $\theta^{(1)}_{N}$, *and* $\widehat{\theta}$. *Which step was better? Explain why, using the two quadratic models of* S *that the methods minimize.*

(e) "Newton converges quadratically, so it is always better." Comment.

Theory needed before solving

For $f_i(\theta) = e^{-\theta x_i}$,

$$J_i = f_i'(\theta) = -x_i e^{-\theta x_i}, \qquad f_i''(\theta) = x_i^2 e^{-\theta x_i}, \qquad r_i = y_i - f_i.$$

The scalar GN and Newton steps are

$$\theta^{\mathrm{new}}_{\mathrm{GN}} = \theta + \frac{\sum_i J_i r_i}{\sum_i J_i^2}, \qquad \theta^{\mathrm{new}}_{N} = \theta + \frac{\sum_i J_i r_i}{\sum_i J_i^2 - \sum_i r_i f_i''}.$$

GN minimizes the locally linearized-model criterion $\|r - J\delta\|^2$, while Newton minimizes the second-order Taylor approximation to S using the full Hessian. A better local Taylor model need not produce a better full step when the step is too large for the approximation to remain accurate.

Detailed solution

(a) At $\theta^{(0)} = 1$,

$$f_1 = e^{-1} = 0.3678794, \qquad f_2 = e^{-2} = 0.1353353.$$

The residuals are observed minus fitted:

$$r_1 = 0.62 - e^{-1} = 0.2521206, \qquad r_2 = 0.18 - e^{-2} = 0.0446647.$$

The first derivatives are

$$J_1 = -1e^{-1} = -0.3678794, \qquad J_2 = -2e^{-2} = -0.2706706,$$

and the second derivatives are

$$f_1'' = 1^2 e^{-1} = 0.3678794, \qquad f_2'' = 2^2 e^{-2} = 0.5413411.$$

(b) First calculate the numerator common to both steps:

$$\begin{aligned} A = \sum_i J_i r_i &= (-0.3678794)(0.2521206) + (-0.2706706)(0.0446647) \\ &\approx -0.1048394. \end{aligned}$$

The GN denominator is

$$B = \sum_i J_i^2 = (0.3678794)^2 + (0.2706706)^2 \approx 0.2085978.$$

Therefore

$$\theta_{\mathrm{GN}}^{(1)} = 1 + \frac{A}{B} = 1 - \frac{0.1048394}{0.2085978} \approx \boxed{0.497409}.$$

(c) For Newton, also calculate the residual-curvature term:

$$\sum_i r_i f_i'' = (0.2521206)(0.3678794) + (0.0446647)(0.5413411) \approx 0.1169288.$$

Thus the Newton denominator is

$$B - \sum_i r_i f_i'' \approx 0.2085978 - 0.1169288 = 0.0916690.$$

So

$$\theta_N^{(1)} = 1 + \frac{-0.1048394}{0.0916690} \approx \boxed{-0.143673}.$$

The full-Hessian correction nearly halves the denominator relative to GN, producing a much larger, negative update.

(d) The objective is

$$S(\theta) = (0.62 - e^{-\theta})^2 + (0.18 - e^{-2\theta})^2.$$

Substituting the parameter values gives:

Parameter value	$S(\theta)$
$\theta^{(0)} = 1$	0.0655597
$\theta_{\mathrm{GN}}^{(1)} \approx 0.497409$	0.0361620
$\theta_N^{(1)} \approx -0.143673$	1.6148427
$\widehat{\theta} \approx 0.6489$	0.01816

Thus the GN step is better for this particular first iteration: it reduces S from about 0.06556 to 0.03616, while the full Newton step increases it dramatically to about 1.615. The given minimizer has a still smaller objective, about 0.01816.

Why? For a parameter increment δ, GN approximates

$$S(\theta + \delta) \approx \|r - J\delta\|^2 = S(\theta) - 2\delta J^T r + \delta^2 J^T J,$$

whereas Newton uses

$$S(\theta+\delta) \approx S(\theta) + S'(\theta)\delta + \tfrac{1}{2}S''(\theta)\delta^2,$$

with $S''/2 = \sum J_i^2 - \sum r_i f_i''$. Here the Newton quadratic predicts a much larger step because its curvature is smaller. But the actual objective is not well represented by that quadratic over such a long step; the trial point crosses into a region where the residuals become large. A line search or damping would reject/shrink that step.

(e) The statement is false. Newton's method can converge quadratically when it is sufficiently close to a regular root/minimizer and the Hessian behaves well. This is a local rate statement, not a guarantee that every full step reduces the objective or that every initial point lies in the convergence region. Here the Newton step is worse than GN because the local quadratic approximation is inaccurate at the proposed step length. Damping, line searches, trust regions, or a safer initial value can make Newton-type algorithms more reliable.

11 Problem 11 – How fast is Gauss–Newton?

Question (transcribed from PS4)

Problem 11. (How fast is Gauss–Newton?) *For a one-parameter model, Gauss–Newton is the iteration $\theta^{(k+1)} = G(\theta^{(k)})$ with*

$$G(\theta) = \theta + \frac{\sum_i J_i(\theta) r_i(\theta)}{\sum_i J_i(\theta)^2}.$$

(a) Show that

$$G'(\widehat{\theta}) = \frac{\sum_i f_i''(\widehat{\theta}) r_i(\widehat{\theta})}{\sum_i J_i(\widehat{\theta})^2}.$$

(b) Deduce the local rate of convergence. When is it faster than linear?

(c) What does the sign of $G'(\widehat{\theta})$ tell you about the iterates?

(d) For the data of Problem 10, compute $G'(\widehat{\theta})$ and compare it with the observed error ratios e_{k+1}/e_k, where $e_k = \theta^{(k)} - \widehat{\theta}$:

$$0.041, \quad -0.083, \quad -0.0895, \quad -0.0890, \quad -0.0890.$$

(e) Can Gauss–Newton fail to converge even when started very close to $\widehat{\theta}$?

Theory needed before solving

For a fixed-point iteration $\theta_{k+1} = G(\theta_k)$ with $G(\widehat{\theta}) = \widehat{\theta}$, Taylor expansion near the fixed point gives

$$e_{k+1} = G'(\widehat{\theta}) e_k + O(e_k^2), \qquad e_k = \theta_k - \widehat{\theta}.$$

If $0 < |G'(\widehat{\theta})| < 1$, convergence is locally linear with asymptotic error ratio $|G'(\widehat{\theta})|$. A negative derivative makes successive errors alternate signs; a positive derivative usually makes them retain their sign locally. If $G'(\widehat{\theta}) = 0$, the linear term vanishes and convergence is faster than linear (typically quadratic for a smooth iteration with nonzero second derivative). If $|G'| > 1$, the fixed point is locally repelling.

Detailed solution

(a) Define

$$A(\theta) = \sum_i J_i r_i, \qquad B(\theta) = \sum_i J_i^2, \quad G(\theta) = \theta + \frac{A(\theta)}{B(\theta)}.$$

At an interior least-squares minimizer, the first-order condition is $S'(\widehat{\theta}) = 0$, equivalently $A(\widehat{\theta}) = \sum_i J_i r_i = 0$. Differentiate each residual and derivative:

$$J_i' = f_i'', \qquad r_i' = -J_i.$$

Therefore

$$A'(\theta) = \sum_i \{J_i' r_i + J_i r_i'\} = \sum_i f_i'' r_i - \sum_i J_i^2 = C(\theta) - B(\theta),$$

where $C(\theta) = \sum_i f_i'' r_i$. By the quotient rule,

$$G'(\theta) = 1 + \frac{A'B - AB'}{B^2}.$$

At $\theta = \widehat{\theta}$, $A = 0$, so

$$G'(\widehat{\theta}) = 1 + \frac{A'(\widehat{\theta})}{B(\widehat{\theta})} = 1 + \frac{C - B}{B} = \boxed{\frac{C}{B} = \frac{\sum_i f_i''(\widehat{\theta}) r_i(\widehat{\theta})}{\sum_i J_i(\widehat{\theta})^2}.}$$

(b) Apply the fixed-point error expansion from the theory section:

$$e_{k+1} = G'(\widehat{\theta}) e_k + O(e_k^2).$$

When $0 < |G'(\widehat{\theta})| < 1$, the ratio e_{k+1}/e_k tends to $G'(\widehat{\theta})$ and the convergence is linear. It is faster than linear when the coefficient of e_k vanishes, namely when $G'(\widehat{\theta}) = 0$; under standard smoothness this gives at least quadratic local convergence. If $|G'(\widehat{\theta})| > 1$, a sufficiently small error is amplified rather than reduced.

(c) The sign tells how the error behaves locally. If $G'(\widehat{\theta}) > 0$, the next error has approximately the same sign as the current error. If $G'(\widehat{\theta}) < 0$, it has the opposite sign, so iterates oscillate around the solution. The magnitude controls the asymptotic linear rate; a negative sign is not itself a failure if the magnitude is below one.

(d) For the Problem 10 model, $f_i = e^{-\theta x_i}$, $J_i = -x_i e^{-\theta x_i}$, and $f_i'' = x_i^2 e^{-\theta x_i}$. The minimizer is approximately $\widehat{\theta} = 0.6488712$. At that value,

$$\begin{aligned}(f_1, f_2) &\approx (0.5226354, 0.2731477),\\ (r_1, r_2) &\approx (0.0973646, -0.0931477),\\ (J_1, J_2) &\approx (-0.5226354, -0.5462955),\\ (f_1'', f_2'') &\approx (0.5226354, 1.0925910).\end{aligned}$$

Thus

$$\sum_i f_i'' r_i \approx (0.5226354)(0.0973646) + (1.0925910)(-0.0931477) \approx -0.0508862,$$

and, using the unrounded values in the ratio,

$$\sum_i J_i^2 \approx 0.5226354^2 + 0.5462955^2 \approx 0.571587, \qquad \boxed{G'(\widehat{\theta}) \approx -0.089026.}$$

The observed error ratios supplied in the question settle near -0.0890, exactly as the local theory predicts. The negative sign explains the alternating errors, and the magnitude 0.089 means that after the iterates enter the local region, the error shrinks by roughly a factor of eleven at each step.

(e) Yes. The local convergence condition is $|G'(\widehat{\theta})| < 1$, not merely that the initial estimate be close in an informal absolute sense. If $|G'| > 1$, the fixed point is locally repelling; if the denominator $\sum J_i^2$ is very small, the map can be unstable; and constraints, singular Jacobians, or a poor local model can disrupt the iteration. Therefore GN can fail even from a close start when the local stability conditions fail or the model is ill-conditioned. In the Problem 10 example, however, $|G'(\widehat{\theta})| \approx 0.089 < 1$, so sufficiently close starts converge locally.

12 Problem 12 – Three tests of one hypothesis

Question (transcribed from PS4)

Problem 12. (Three tests of one hypothesis) *Theophylline data, subject 1 ($n = 11$ observations, $p = 3$ parameters; use the t_8 distribution). The one-compartment fit, parametrized by $(\log k_e, \log k_a, \log CL)$, gives*

$$\widehat{\log k_e} = -2.9196, \qquad \mathrm{se}(\widehat{\log k_e}) = 0.1709, \qquad S(\widehat{\theta}) = 4.2860.$$

We test $H_0 : k_e = 0.08$ per hour; equivalently, $\log k_e = \log 0.08$; equivalently, the half-life is $t_{1/2} = \log 2/0.08 = 8.66$ hours. Use $t_{8,0.975} = 2.306$ and $F_{1,8,0.95} = 5.318$.

(a) Compute the Wald t statistic on the log k_e scale, and state your decision at the 5% level.

(b) Use the delta method to obtain $\widehat{k}_e$, its standard error, and the Wald test on the k_e scale.

(c) Do the same on the half-life scale.

(d) Show the exact identity $t_{\text{half-life scale}} = -t_{k_e \text{ scale}}$, where $k_0 = 0.08$.

(e) Refitting with k_e fixed at 0.08 gives $S_R = 7.5678$. Compute the extra-sum-of-squares F statistic. Why can its value not depend on the scale?

Theory needed before solving

A Wald test standardizes the estimate minus the null value by an estimated standard error. Under a smooth reparameterization $\phi = g(\theta)$, the delta method gives $\mathrm{se}(\widehat{\phi}) \approx |g'(\widehat{\theta})|\,\mathrm{se}(\widehat{\theta})$. Wald statistics are generally not invariant to nonlinear transformations, because a symmetric local approximation on one scale need not be symmetric on another. By contrast, a likelihood-ratio or extra-sum-of-squares test depends on the fitted models and their objective values; the null constraint represents the same subset of the model space after a one-to-one reparameterization, so the objective comparison is invariant to the scale used to label the constrained parameter.

Detailed solution

(a) The null value on the log scale is

$$\log k_0 = \log(0.08) \approx -2.5257286.$$

The Wald statistic is

$$t_{\log k_e} = \frac{-2.9196 - \log(0.08)}{0.1709} \approx \boxed{-2.30469}.$$

Since $|t| = 2.30469 < 2.306$, using the rounded values in the question we just fail to reject H_0 at the 5% two-sided level. This is an extremely borderline result; the displayed data are rounded, so the comparison should not be interpreted as high-precision evidence.

(b) Transform the estimate back:

$$\widehat{k}_e = e^{-2.9196} \approx \boxed{0.0539553 \text{ hour}^{-1}}.$$

For $g(\theta) = e^{\theta}$, $g'(\theta) = e^{\theta}$. The delta-method standard error is therefore

$$\operatorname{se}(\widehat{k}_e) \approx \widehat{k}_e \operatorname{se}(\widehat{\log k_e}) = (0.0539553)(0.1709) \approx \boxed{0.0092210}.$$

The Wald statistic on the k_e scale is

$$t_{k_e} = \frac{0.0539553 - 0.08}{0.0092210} \approx \boxed{-2.82452}.$$

Now $|t| > 2.306$, so the delta-method Wald test rejects H_0 at the 5% level on this scale.

(c) Let $h = t_{1/2} = \log 2/k_e$. Under the estimate,

$$\widehat{h} = \frac{\log 2}{\widehat{k}_e} \approx \boxed{12.8467 \text{ hours}}.$$

The null half-life is $h_0 = \log 2/0.08 \approx 8.66434$ hours. As a function of $\gamma = \log k_e$, $h(\gamma) = \log 2\, e^{-\gamma}$, whose derivative is $-h(\gamma)$. Thus the delta-method standard error is

$$\operatorname{se}(\widehat{h}) \approx \widehat{h} \operatorname{se}(\widehat{\log k_e}) = (12.8467)(0.1709) \approx \boxed{2.19550 \text{ hours}}.$$

The half-life scale Wald statistic is

$$t_h = \frac{12.8467 - 8.66434}{2.19550} \approx \boxed{1.90497}.$$

Since $|t_h| < 2.306$, this Wald test does not reject at 5%.

(d) To verify the identity, write $\widehat{k} = \widehat{k}_e$ and $k_0 = 0.08$. Delta-method standard errors imply

$$t_{k_e} = \frac{\widehat{k} - k_0}{\widehat{k}\, s_{\log k}}, \qquad s_{\log k} = 0.1709.$$

For half-life,

$$t_h = \frac{\log 2/\widehat{k} - \log 2/k_0}{(\log 2/\widehat{k}) s_{\log k}} = \frac{1 - \widehat{k}/k_0}{s_{\log k}}.$$

On the other hand,

$$-t_{k_e} \frac{\widehat{k}}{k_0} = -\frac{\widehat{k} - k_0}{\widehat{k} s_{\log k}} \frac{\widehat{k}}{k_0} = \frac{1 - \widehat{k}/k_0}{s_{\log k}} = t_h.$$

Thus the exact algebraic identity for these delta-method Wald statistics is

$$\boxed{t_h = -t_{k_e} \frac{\widehat{k}_e}{k_0}.}$$

Numerically, $-(-2.82452)(0.0539553/0.08) \approx 1.90497$.

(e) The full model has $S_F = S(\widehat{\theta}) = 4.2860$ and residual degrees of freedom $n - p = 8$. Fixing one parameter imposes one restriction, so $q = 1$. The extra-sum-of-squares statistic is

$$F = \frac{(S_R - S_F)/q}{S_F/(n-p)} = \frac{7.5678 - 4.2860}{4.2860/8} = \boxed{6.12562}.$$

Since $6.12562 > F_{1,8,0.95} = 5.318$, the extra-sum-of-squares test rejects the null at the 5% level.

Why is this test invariant to the scale? The restrictions $k_e = 0.08$, $\log k_e = \log 0.08$, and $h = \log 2/0.08$ describe the same constrained physical model. Refitting under any of those equivalent descriptions gives the same restricted fitted mean and hence the same restricted RSS S_R; the unrestricted fit also has the same RSS. The F statistic uses only those RSS values and degrees of freedom. The three Wald conclusions differ because each Wald test uses a local symmetric approximation in its chosen scale; the extra-sum-of-squares test does not.

13 Problem 13 – Why profile equals Wald in linear models

Question (transcribed from PS4)

Problem 13. (Why profile equals Wald in linear models)

(a) In the linear model $y = X\beta + \varepsilon$ (X of full column rank), show that for every b,

$$\widetilde{S}(b) - S(\widehat{\beta}) = \frac{(b - \widehat{\beta}_j)^2}{[(X^T X)^{-1}]_{jj}}, \qquad \widetilde{S}(b) = \min_{\beta:\, \beta_j = b} S(\beta).$$

Deduce that the profile t function

$$\tau(b) = \operatorname{sign}(b - \widehat{\beta}_j) \frac{\sqrt{\widetilde{S}(b) - S(\widehat{\beta})}}{s}$$

equals $(b - \widehat{\beta}_j)/\operatorname{se}(\widehat{\beta}_j)$, a straight line.

(b) In a nonlinear model, let $\phi = g(\theta_j)$ with g strictly increasing. Show that the profile interval for ϕ is exactly g(profile interval for θ_j). Is the same true for Wald intervals?

(c) Explain why curvature in the plot of $\tau(\theta_j)$ against θ_j is a diagnostic of non-linearity.

Theory needed before solving

In a full-rank linear model, the least-squares residual is orthogonal to the columns of X. The residual-sum-of-squares identity is

$$S(\beta) = S(\widehat{\beta}) + (\beta - \widehat{\beta})^T X^T X (\beta - \widehat{\beta}).$$

A profile objective fixes the parameter of interest and minimizes over all nuisance parameters. A confidence set based on a profile objective is defined by the objective values themselves; a one-to-one monotone reparameterization simply relabels the same constrained models. A Wald interval, however, uses an estimate and a local standard error and is usually symmetric on whichever scale it is constructed, so it is not generally invariant under nonlinear transformations.

Detailed solution

(a) Let $d = \beta - \widehat{\beta}$. Since X has full column rank,

$$S(\beta) = \|y - X\beta\|^2 = S(\widehat{\beta}) + d^T X^T X d,$$

because $y - X\widehat{\beta}$ is orthogonal to Xd. The constraint $\beta_j = b$ is $e_j^T \beta = b$, so $e_j^T d = b - \widehat{\beta}_j =: c$. We must minimize $d^T A d$ subject to $e_j^T d = c$, where $A = X^T X$ is positive definite.

Use a Lagrange multiplier:

$$\mathcal{L}(d, \lambda) = d^T A d - 2\lambda(e_j^T d - c).$$

Differentiation gives $2Ad - 2\lambda e_j = 0$, so $d = \lambda A^{-1} e_j$. Enforcing the constraint,

$$c = e_j^T d = \lambda e_j^T A^{-1} e_j = \lambda [(X^T X)^{-1}]_{jj}.$$

Therefore $\lambda = c/[(X^T X)^{-1}]_{jj}$ and the minimized quadratic penalty is

$$d^T A d = \lambda^2 e_j^T A^{-1} e_j = \frac{c^2}{[(X^T X)^{-1}]_{jj}}.$$

Substituting $c = b - \widehat{\beta}_j$ proves

$$\boxed{\widetilde{S}(b) - S(\widehat{\beta}) = \frac{(b - \widehat{\beta}_j)^2}{[(X^T X)^{-1}]_{jj}}.}$$

With $s^2 = S(\widehat{\beta})/(n-p)$, the usual standard error is $\mathrm{se}(\widehat{\beta}_j) = s\sqrt{[(X^T X)^{-1}]_{jj}}$. Hence

$$\begin{aligned}\tau(b) &= \mathrm{sign}(b - \widehat{\beta}_j) \frac{|b - \widehat{\beta}_j| / \sqrt{[(X^T X)^{-1}]_{jj}}}{s} \\ &= \frac{b - \widehat{\beta}_j}{s\sqrt{[(X^T X)^{-1}]_{jj}}} = \boxed{\frac{b - \widehat{\beta}_j}{\mathrm{se}(\widehat{\beta}_j)}}.\end{aligned}$$

This is a straight line in b, passing through zero at the estimate.

(b) Let C_θ be the profile confidence interval for θ_j. If g is strictly increasing, it is one-to-one on its domain. Testing or profiling a candidate value ϕ is exactly the same constrained fit as setting $\theta_j = g^{-1}(\phi)$. Thus the set of accepted values on the ϕ scale is the image of the accepted set on the θ_j scale:

$$\boxed{C_\phi = g(C_\theta).}$$

For an interval $C_\theta = [L, U]$, strict monotonicity gives $C_\phi = [g(L), g(U)]$.

Wald intervals do not generally have this exact transformation property. On the θ_j scale, a Wald interval is $\widehat{\theta}_j \pm z\,\mathrm{se}(\widehat{\theta}_j)$; on the transformed scale it is typically $g(\widehat{\theta}_j) \pm z\,|g'(\widehat{\theta}_j)|\,\mathrm{se}(\widehat{\theta}_j)$. That symmetric interval is not generally equal to $[g(L), g(U)]$. Equality holds for affine transformations (and may hold approximately for nearly linear g over a narrow interval), but not for a general nonlinear transformation.

(c) In a linear model, part (a) proves the profile t curve is exactly a straight line. In a nonlinear model, the profiled objective need not be a quadratic function of θ_j; its profile curve and the corresponding signed square-root statistic can bend. Such curvature indicates that a local linear approximation is not describing the likelihood/RSS geometry equally well in both directions. It warns that Wald intervals and tests may be parameterization-sensitive, asymmetric, or materially different from profile-based inference. Curvature is a diagnostic rather than an absolute proof on its own: it should be interpreted alongside identifiability, sample size, and model fit.

14 Problem 14 – Where should we measure?

Question (transcribed from PS4)

Problem 14. (Where should we measure?)

(a) Model $\theta_1 e^{-\theta_2 x}$ ($\theta_1, \theta_2 > 0$), with $n/2$ observations at $x = 0$ and $n/2$ at $x = c > 0$. Show that

$$\mathrm{Var}(\widehat{\theta}_2) \approx \frac{2\sigma^2\theta_2^2}{n\theta_1^2} \cdot \frac{e^{2u}+1}{u^2}, \qquad u = \theta_2 c,$$

and that the best c solves $u = 1 + e^{-2u}$. Solve this equation by the fixed-point iteration $u^{(k+1)} = 1 + e^{-2u^{(k)}}$ from $u^{(0)} = 1$ (three or four steps suffice), and express the optimal c in half-lives. Why can we not use this design directly?

(b) Michaelis–Menten $V_m x/(K+x)$, with $n/2$ observations at a and $n/2$ at b, where $0 < a < b$ and b is the largest feasible concentration. Show that $\det(J^T J) \propto \{a(b-a)/(K+a)^2\}^2$, and that the D-optimal choice (maximizing $\det(J^T J)$) is

$$a^* = \frac{Kb}{2K+b}.$$

Evaluate it for $K = 0.0641$, $b = 1.10$ (Puromycin). What happens as $a \to b$? As $a \to 0$?

Theory needed before solving

For nonlinear least squares with independent variance σ^2, the local covariance approximation is $\sigma^2(J^T J)^{-1}$. An optimal design chooses the predictor values to make the information matrix $J^T J$ informative. For a scalar target parameter, one can minimize its approximate variance directly. For multiple parameters, D-optimality maximizes $\det(J^T J)$, which minimizes the volume of the local confidence ellipsoid (equivalently maximizes the determinant of the local information matrix). Such designs often depend on unknown model parameters, making them locally optimal rather than universally optimal.

Detailed solution

(a) The mean is $f(x) = \theta_1 e^{-\theta_2 x}$, so the Jacobian row at design value x is

$$J(x) = \left(e^{-\theta_2 x},\ -\theta_1 x e^{-\theta_2 x} \right).$$

There are $n/2$ observations at each of 0 and c. Put $u = \theta_2 c$. The information matrix entries are

$$A = \sum_i \left(\frac{\partial f_i}{\partial \theta_1} \right)^2 = \frac{n}{2}(1 + e^{-2u}),$$

$$B = \sum_i \frac{\partial f_i}{\partial \theta_1} \frac{\partial f_i}{\partial \theta_2} = -\frac{n}{2}\theta_1 c e^{-2u},$$

$$D = \sum_i \left(\frac{\partial f_i}{\partial \theta_2} \right)^2 = \frac{n}{2}\theta_1^2 c^2 e^{-2u}.$$

Thus

$$J^T J = \begin{pmatrix} A & B \\ B & D \end{pmatrix}, \qquad \det(J^T J) = AD - B^2 = \frac{n^2}{4}\theta_1^2 c^2 e^{-2u}.$$

For a 2×2 matrix, the bottom-right entry of the inverse is $A/\det(J^TJ)$. Therefore

$$\begin{aligned}\mathrm{Var}(\widehat{\theta}_2) &\approx \sigma^2[(J^TJ)^{-1}]_{22} \\ &= \sigma^2 \frac{\frac{n}{2}(1+e^{-2u})}{\frac{n^2}{4}\theta_1^2 c^2 e^{-2u}} \\ &= \frac{2\sigma^2}{n\theta_1^2 c^2}(e^{2u}+1).\end{aligned}$$

Since $c = u/\theta_2$, this is exactly

$$\boxed{\mathrm{Var}(\widehat{\theta}_2) \approx \frac{2\sigma^2\theta_2^2}{n\theta_1^2}\frac{e^{2u}+1}{u^2}.}$$

The constant factor does not depend on u, so minimize $q(u) = (e^{2u}+1)/u^2$ for $u > 0$. Differentiating,

$$q'(u) = \frac{2ue^{2u} - 2(e^{2u}+1)}{u^3}.$$

Setting the numerator to zero gives

$$ue^{2u} = e^{2u}+1 \quad\Longleftrightarrow\quad (u-1)e^{2u} = 1 \quad\Longleftrightarrow\quad \boxed{u = 1+e^{-2u}}.$$

Use the requested fixed-point iteration starting with $u^{(0)} = 1$:

k	$u^{(k)}$
0	1.000000
1	$1+e^{-2} = 1.135335$
2	$1+e^{-2(1.135335)} \approx 1.103243$
3	$1+e^{-2(1.103243)} \approx 1.110087$
4	$1+e^{-2(1.110087)} \approx 1.108590$

So $u^* \approx 1.1086$ (the fixed point is approximately 1.1089 to four significant figures). The half-life under exponential decay is $t_{1/2} = \log 2/\theta_2$. Since $c = u/\theta_2$,

$$\frac{c^*}{t_{1/2}} = \frac{u^*/\theta_2}{\log 2/\theta_2} = \frac{u^*}{\log 2} \approx \boxed{1.60 \text{ half-lives}}.$$

We cannot use the design directly because the optimal value is $c^* = u^*/\theta_2$, and θ_2 is unknown before data are collected. A pilot study, prior scientific knowledge, or sequential/adaptive design can supply an initial rate estimate; then the design can be updated. This is a locally optimal design because the calculation uses the unknown true parameters in the local covariance approximation.

(b) Let $m = n/2$. For each design value x, the Jacobian row with respect to (V_m, K) is

$$j(x) = \left(\frac{x}{K+x},\ -\frac{V_m x}{(K+x)^2}\right).$$

There are m repeated rows $j(a)$ and m rows $j(b)$, so

$$J^TJ = m\{j(a)^Tj(a) + j(b)^Tj(b)\}.$$

For two row types, the determinant of this 2×2 information matrix is m^2 times the square of the determinant formed from one row of each type:

$$\det(J^TJ) = m^2 \det\begin{pmatrix} j(a) \\ j(b)\end{pmatrix}^2.$$

Calculate the row determinant:

$$\det\begin{pmatrix} j(a) \\ j(b) \end{pmatrix} = \frac{a}{K+a}\left(-\frac{V_m b}{(K+b)^2}\right) - \frac{b}{K+b}\left(-\frac{V_m a}{(K+a)^2}\right)$$
$$= \frac{V_m ab(b-a)}{(K+a)^2(K+b)^2}.$$

Thus

$$\det(J^T J) = m^2 \frac{V_m^2 a^2 b^2 (b-a)^2}{(K+a)^4(K+b)^4} \propto \left\{\frac{a(b-a)}{(K+a)^2}\right\}^2,$$

where the proportionality constant does not depend on the choice of a. Hence maximize

$$q(a) = \frac{a(b-a)}{(K+a)^2}, \qquad 0 < a < b.$$

Since $q(a) > 0$ on the interior, maximize its logarithm:

$$\frac{d}{da}\log q(a) = \frac{1}{a} - \frac{1}{b-a} - \frac{2}{K+a} = 0.$$

Combining terms gives

$$\frac{b-2a}{a(b-a)} = \frac{2}{K+a} \quad \Longrightarrow \quad (b-2a)(K+a) = 2a(b-a).$$

After expansion and cancellation, $bK = a(2K+b)$. Therefore

$$\boxed{a^* = \frac{Kb}{2K+b}.}$$

For $K = 0.0641$ and $b = 1.10$,

$$a^* = \frac{(0.0641)(1.10)}{2(0.0641)+1.10} = \frac{0.07051}{1.2282} \approx \boxed{0.05741}.$$

As $a \to b$, the two design points coalesce and $b - a \to 0$, so $\det(J^T J) \to 0$. The design then supplies essentially one distinct concentration and cannot separate both parameters well. As $a \to 0$, the determinant also tends to zero because both derivatives at $x = 0$ are zero; near zero, $f(x) \approx (V_m/K)x$, so the data mainly identify the ratio V_m/K rather than V_m and K separately. The optimal a^* balances the need for low-concentration information about the initial slope against the need for concentration separation to learn the saturation parameter.

Consolidated numerical answers

This table is a quick-check summary; the derivations and qualifications are in the corresponding problem sections.

Problem	Key result(s)
1	(a) linear; (b) power law log-linear only under compatible multiplicative error; (c) conditionally linear; (d) nonlinear constrained quadratic with sign non-identifiability; (e) Michaelis–Menten intrinsically nonlinear.
2	(a) only $\theta_1 e^{\theta_2}$ and θ_3; (b1) one mean combination only; (b2) identifiable with two distinct positive design values, but NLSE need not exist; (c) label switching; (d) frequency aliases.
3	$\widehat{\beta} = 57/28 = 2.035714$, $\widehat{\theta} \approx 1.267376$; from $\theta^{(0)} = 1$, first two GN iterates 1.345238, 1.271796.
4	Nonexistence probability for $\theta = 0, \sigma = 2, n = 4$: 0.1587; for $e^{\theta} = 5$, bias ≈ -0.0200, second-order SD ≈ 0.2020.
5	(b) $(\widehat{\theta}_1, \widehat{\theta}_2) = (100, 0.458145\ \mathrm{h}^{-1})$; (c) on the two group means, $\widehat{V}_m \approx 212.590$, $\widehat{K} \approx 0.049135$.
6	Coffee: $\widehat{\alpha} \approx 23.333°\mathrm{C}$, $\widehat{k} \approx 0.071334\ \mathrm{min}^{-1}$, $\widehat{y}(15) = 46.2°\mathrm{C}$. BOD days 1–3 invalid for $k > 0$; days 3–5 yield $\widehat{\alpha} \approx 15.5385$, $\widehat{k} \approx 2.01490$.
7	$\widehat{\theta}_1(\theta_2) = y^T g/(g^T g)$ and $\widetilde{S} = \|y\|^2 - (y^T g)^2/\|g\|^2$.
8	$\widehat{\beta}_0 \approx -3.31419$, $\widehat{\beta}_1 \approx 1.97918$, odds ratio ≈ 7.23684, 95% Wald OR interval $\approx (1.78, 29.41)$.
9	$\ell = \sum_i [y_i x_i^T \beta - e^{x_i^T \beta} - \log(y_i!)]$; IRLS $w_i = \mu_i$, $z_i = \eta_i + (y_i - \mu_i)/\mu_i$; first step $\beta^{(1)} \approx (0.75383, 0.54545)$.
10	GN step ≈ 0.497409, Newton step ≈ -0.143673; objective values 0.036162 and 1.614843, respectively.
11	For Problem 10, $G'(\widehat{\theta}) \approx -0.089026$, matching the observed error-ratio limit -0.0890.
12	Wald t statistics: log scale -2.30469 (do not reject, borderline), k_e scale -2.82452 (reject), half-life scale 1.90497 (do not reject); extra-SS $F \approx 6.12562$ (reject).
13	In a linear model, $\tau(b) = (b - \widehat{\beta}_j)/\operatorname{se}(\widehat{\beta}_j)$ exactly. Monotone profile intervals transform exactly; nonlinear Wald intervals generally do not.
14	(a) $u^* \approx 1.1089$, so the optimal c is about 1.60 half-lives; (b) $a^* \approx 0.05741$.

Final checklist for revision

Before using an answer in an assignment or exam, check the following conceptual distinctions:

- **Model linearity versus curve shape:** polynomial, log-x, and known-frequency sine/cosine regressors can be curved in x while remaining linear in coefficients.
- **Identifiability versus convergence:** a unique statistical estimate may exist even if a particular iterative algorithm fails; conversely, an optimizer may converge to a point while parameters are not identifiable.
- **Algebraic linearization versus statistical equivalence:** logs and reciprocals alter the error structure. A straight-line identity does not automatically justify ordinary least squares on transformed observations.
- **GN versus Newton:** GN uses $J^T J$ curvature and can be stable when a Newton step overshoots; Newton's local quadratic convergence is not a global guarantee.
- **Wald versus objective-based tests:** Wald tests can change under nonlinear reparameterization, whereas likelihood-ratio and extra-sum-of-squares tests compare the same underlying

constrained and unconstrained fits.

- **Local design optimality:** formulas for optimal design often depend on unknown true parameters and may require pilot estimates or sequential refinement.